Year 9 · Chapter 11 · Coordinate geometry and graphs: answers (teacher copy)

11.1 The length of an interval (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: $d = \sqrt{((4-1)^2 + (5-1)^2)} = \sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$
2. Ex 1b: $d = \sqrt{((2-(-2))^2 + (-1-2)^2)} = \sqrt{(4^2 + (-3)^2)} = \sqrt{25} = 5$
3. Ex 1c: $d = \sqrt{((3-(-1))^2 + (-1-2)^2)} = \sqrt{(4^2 + (-3)^2)} = \sqrt{25} = 5$
4. Ex 1d: $d = \sqrt{((4-1)^2 + (4-(-2))^2)} = \sqrt{(3^2 + 6^2)} = \sqrt{45} = 6.7$
5. Ex 2a: $d = \sqrt{((4-(-4))^2 + (5-(-1))^2)} = \sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$
6. Ex 2b: $d = \sqrt{((3-(-2))^2 + (2-(-3))^2)} = \sqrt{(5^2 + 5^2)} = \sqrt{50} = 7.1$
7. Ex 2c: $d = \sqrt{((1-(-5))^2 + (-4-4)^2)} = \sqrt{(6^2 + (-8)^2)} = \sqrt{100} = 10$
8. Ex 2d: $d = \sqrt{((-3-2)^2 + (3-(-4))^2)} = \sqrt{((-5)^2 + 7^2)} = \sqrt{74} = 8.6$

Easy

1. 5
2. 10
3. 5
4. 10
5. $\sqrt{50} = 7.1$
6. $\sqrt{61} = 7.8$
7. $\sqrt{65} = 8.1$
8. $\sqrt{89} = 9.4$
9. 5
10. 10
11. $\sqrt{61} = 7.8$
12. $\sqrt{65} = 8.1$
13. 9
14. 8

Medium

1. 10
2. P(−3, 2), Q(4, −1) (Year 8)
3. $\sqrt{65} = 8.1$
4. $\sqrt{89} = 9.4$
5. $6 \times 4 = 24$ square units (10.4)
6. $\sqrt{72} = 8.5$

Worded problems

1. $d = \sqrt{((5-(-3))^2 + (8-2)^2)} = \sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$ units, so 1 000 m
2. $d = \sqrt{((14-2)^2 + (6-1)^2)} = \sqrt{(12^2 + 5^2)} = \sqrt{169} = 13$ km; road $12 + 5 = 17$ km, so $17 - 13 = 4$ km shorter
3. $AC = \sqrt{(3^2 + 4^2)} = 5$, $BC = \sqrt{((-3)^2 + 4^2)} = 5$, AB = 6: AC = BC, so isosceles. $P = 5 + 5 + 6 = 16$ units

11.2 The midpoint of an interval

(booklet p. 7)

WE DO examples (teacher)

1. Ex 1a: $M = ((1+5) \div 2, (1+3) \div 2) = (3, 2)$
2. Ex 1b: $M = ((-4+2) \div 2, (3+(-1)) \div 2) = (-1, 1)$
3. Ex 1c: $M = ((-3+3) \div 2, (-4+2) \div 2) = (0, -1)$
4. Ex 1d: $M = ((-1+4) \div 2, (4+(-3)) \div 2) = (1.5, 0.5)$
5. Ex 2a: $M = ((-4+4) \div 2, (-1+5) \div 2) = (0, 2)$
6. Ex 2b: $M = ((-3+4) \div 2, (3+(-4)) \div 2) = (0.5, -0.5)$
7. Ex 2c: $M = ((-5+1) \div 2, (-2+4) \div 2) = (-2, 1)$
8. Ex 2d: $x: 2 \times 1 - (-3) = 5$, $y: 2 \times (-1) - 2 = -4$, so B = (5, −4)

Easy

1. (3, 3)
2. (−1, 1)
3. (−1, 3)
4. (−2, −2)
5. (0.5, −0.5)
6. (−1.5, 1.5)
7. (−0.5, −1)
8. (0.5, −1)
9. (−1, −1)
10. (1, 2)
11. (−1.5, 1.5)
12. (2.5, −0.5)
13. (0, 1)
14. (2, −1)

Medium

1. (−1, 0)
2. 5 (11.1)
3. (−0.5, 1)
4. (0.5, −1)
5. $2 \times (6 + 5) = 22$ units (10.4)
6. (−1.5, −0.5)

Worded problems

1. $M = ((-4+8) \div 2, (3+(-1)) \div 2) = (2, 1)$
2. Centre: $M = ((-1+7) \div 2, (2+8) \div 2) = (3, 5)$; diameter $d = \sqrt{((7-(-1))^2 + (8-2)^2)} = \sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$, so the radius is 5
3. Tap: $M = ((2+10) \div 2, (1+7) \div 2) = (6, 4)$; $d = \sqrt{((6-2)^2 + (4-1)^2)} = \sqrt{(4^2 + 3^2)} = \sqrt{25} = 5$ m

11.3 The gradient of an interval

(booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: m = (5 − 1) ÷ (3 − 1) = 4 ÷ 2 = 2
2. Ex 1b: m = (1 − 3) ÷ (2 − (−2)) = −2 ÷ 4 = −½
3. Ex 1c: m = (4 − (−4)) ÷ (1 − (−3)) = 8 ÷ 4 = 2
4. Ex 1d: m = 0 ÷ 8 = 0 (horizontal)
5. Ex 2a: m = (3 − (−3)) ÷ (4 − (−4)) = 6 ÷ 8 = ¾
6. Ex 2b: m = (−4 − 4) ÷ (2 − (−2)) = −8 ÷ 4 = −2
7. Ex 2c: m = (3 − (−1)) ÷ (5 − (−5)) = 4 ÷ 10 = ⅖
8. Ex 2d: run = 2 − 2 = 0: the gradient is undefined (vertical)

Easy

1. 2
2. −2
3. 1
4. −1
5. ½
6. −½
7. 3
8. −¾
9. 2
10. −½
11. ¾
12. −2
13. 0
14. undefined (vertical)

Medium

1. 1
2. (0, 1) (11.2)
3. −1
4. ½
5. 5 (11.1)
6. 8/3

Worded problems

1. m = 45 ÷ 600 = 0.075, which is 7.5%
2. Roof: m = 2.4 ÷ 4 = 0.6. Ramp: 0.5. 0.6 > 0.5, so yes, the roof is steeper
3. m(AB) = 4 ÷ 2 = 2 and m(BC) = 6 ÷ 3 = 2: the same gradient through the shared point B, so A, B and C are on one line

11.4 Graphing linear equations (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: y: −3, −1, 1, 3, 5
2. Ex 1b: Plot (−2, −3), (−1, −1), (0, 1), (1, 3), (2, 5) and rule a line
3. Ex 1c: y = 0: 2x + 1 = 0, x = −½ → (−½, 0); x = 0: y = 1 → (0, 1)
4. Ex 1d: 2 × 3 + 1 = 7 ✓ Yes
5. Ex 2a: Table y: 7, 5, 3, 1, −1 (x = −2 to 2); plot and rule the line
6. Ex 2b: y = 0: −2x + 3 = 0, x = 1.5 → (1.5, 0); x = 0: (0, 3)
7. Ex 2c: y = 0: 3x = 6, x = 2 → (2, 0); x = 0: 2y = 6, y = 3 → (0, 3)
8. Ex 2d: x = −2: a vertical line through −2 on the x-axis; y = 3: a horizontal line through 3 on the y-axis

Easy

1. y: 0, 1, 2, 3, 4; x-intercept (−2, 0); y-intercept (0, 2)
2. y: 5, 4, 3, 2, 1; x-intercept (3, 0); y-intercept (0, 3)
3. y: −6, −4, −2, 0, 2; x-intercept (1, 0); y-intercept (0, −2)
4. y: −4, −3, −2, −1, 0; x-intercept (2, 0); y-intercept (0, −2)
5. y: 6, 4, 2, 0, −2; x-intercept (1, 0); y-intercept (0, 2)
6. y: −6, −3, 0, 3, 6; x-intercept (0, 0); y-intercept (0, 0)
7. y: 1, 2, 3, 4, 5; x-intercept (−3, 0); y-intercept (0, 3)
8. y: 0, −1, −2, −3, −4; x-intercept (−2, 0); y-intercept (0, −2)
9. y: 0, 0.5, 1, 1.5, 2; x-intercept (−2, 0); y-intercept (0, 1)
10. y: 3, 2.5, 2, 1.5, 1; x-intercept (4, 0); y-intercept (0, 2)
11. y: −8, −6, −4, −2, 0; x-intercept (2, 0); y-intercept (0, −4)
12. y: 6, 3, 0, −3, −6; x-intercept (0, 0); y-intercept (0, 0)
13. y: 2, 0, −2, −4, −6; x-intercept (−1, 0); y-intercept (0, −2)
14. y: −5, −4, −3, −2, −1; x-intercept (3, 0); y-intercept (0, −3)

Medium

1. (2, 0) and (0, −4)
2. 3/2 (11.3)
3. (3, 0) and (0, 3)
4. (−4, 0) and (0, 2)
5. 5 (11.1)
6. (−3, 0) and (0, −3)

Worded problems

1. C: 40, 100, 160, 220. C-intercept 40: the $40 call-out fee (charged for 0 hours)
2. (6, 0) and (0, 9); d = $\sqrt{(6^2 + 9^2)} = \sqrt{117} = 10.8$
3. Width 5 − (−1) = 6, height 2 − (−3) = 5: P = 22 units, A = 30 square units

11.5 The gradient–intercept equation

(booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: m = 3, c = 1
2. Ex 1b: y = −x + 8: m = −1, c = 8
3. Ex 1c: m = ⅕, c = 4
4. Ex 1d: Plot (0, −3); go across 1, up 2 to (1, −1); rule the line
5. Ex 2a: 2y = −3x + 12, so y = −1.5x + 6: m = −1.5, c = 6
6. Ex 2b: Plot (0, 4); go across 1, down 2 to (1, 2); rule the line
7. Ex 2c: y = −3x + 5
8. Ex 2d: Parallel, so m = 4: y = 4x + 3

Easy

1. m = 2, c = 1: (0, 1) and (1, 3)
2. m = −1, c = 4: (0, 4) and (1, 3)
3. m = 3, c = -2: (0, -2) and (1, 1)
4. m = −2, c = 3: (0, 3) and (1, 1)
5. m = 1, c = -4: (0, -4) and (1, -3)
6. m = ½, c = 2: (0, 2) and (2, 3)
7. m = −1, c = -1: (0, -1) and (1, -2)
8. m = 2, c = 0: (0, 0) and (1, 2)
9. m = −2, c = 4: (0, 4) and (1, 2)
10. m = 3, c = 1: (0, 1) and (1, 4)
11. m = −1, c = -2: (0, -2) and (1, -3)
12. m = ½, c = 2: (0, 2) and (2, 3)
13. m = −2, c = 3: (0, 3) and (1, 1)
14. m = 1, c = -2: (0, -2) and (1, -1)

Medium

1. m = −3, c = 2: (0, 2) and (1, -1)
2. (2, 0) and (0, 4) (11.4)
3. m = −2, c = -1: (0, -1) and (1, -3)
4. m = −½, c = 1: (0, 1) and (2, 0)
5. ½ (11.3)
6. m = 3, c = -2: (0, -2) and (1, 1)

Worded problems

1. Gradient 0.5: $0.50 for each gigabyte. C-intercept 15: the $15 monthly fee (0 GB used)
2. m = 2, c = 3 − 5 = −2: y = 2x − 2. x-intercept: 0 = 2x − 2, x = 1
3. They did not divide every term by 2. 2y = 6x + 8, so y = 3x + 4: m = 3, c = 4

11.6 Finding the equation of a line

(booklet p. 23)

WE DO examples (teacher)

1. Ex 1a: c = 2, m = 2 ÷ 2 = 1: y = x + 2
2. Ex 1b: c = 3, m = −4 ÷ 2 = −2: y = −2x + 3
3. Ex 1c: c = −1, m = 2 ÷ 4 = ½: y = ½x − 1
4. Ex 1d: c = 1, m = −3 ÷ 3 = −1: y = −x + 1
5. Ex 2a: m = −4 ÷ 6 = −⅔, c = 2: y = −⅔x + 2
6. Ex 2b: m = 6 ÷ 2 = 3; −1 = 3 × 1 + c, c = −4: y = 3x − 4
7. Ex 2c: m = −5 ÷ 5 = −1, c = 1: y = −x + 1
8. Ex 2d: m = 2 ÷ 8 = ¼, c = 2: y = ¼x + 2

Easy

1. m = 2, c = 1: y = 2x + 1
2. m = −1, c = 3: y = −x + 3
3. m = 3, c = -2: y = 3x − 2
4. m = −2, c = 1: y = −2x + 1
5. m = ½, c = 1: y = ½x + 1
6. m = 1, c = -3: y = x − 3
7. m = −½, c = 2: y = −½x + 2
8. m = 2, c = 3: y = 2x + 3
9. m = 2, c = -1: y = 2x − 1
10. m = −1, c = 2: y = −x + 2
11. m = ½, c = -1: y = ½x − 1
12. m = −3, c = 2: y = −3x + 2
13. m = 1, c = 1: y = x + 1
14. m = −2, c = -2: y = −2x − 2

Medium

1. m = 4, c = -3: y = 4x − 3
2. m = −2, c = 3: (0, 3) and (1, 1) (11.5)
3. m = −½, c = 3: y = −½x + 3
4. m = ¾, c = 0: y = ¾x
5. (4, 0) and (0, −4) (11.4)
6. m = −3, c = -1: y = −3x − 1

Worded problems

1. m = (18 − 24) ÷ 2 = −3, c = 24: h = −3t + 24. 0 = −3t + 24, so t = 8 hours
2. m = (25 − 13) ÷ (10 − 4) = 2; 13 = 2 × 4 + c, c = 5: C = 2d + 5. Flagfall $5
3. m = 12 ÷ 6 = 2; −1 = 2 × (−2) + c, c = 3: y = 2x + 3. 0 = 2x + 3, x = −1.5: (−1.5, 0)

11.7 Solving linear equations graphically

(booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: The lines meet at x = 2. Check: 2 × 2 + 1 = 5 ✓
2. Ex 1b: x = −2. Check: 2 × (−2) + 1 = −3 ✓
3. Ex 1c: x = 3. Check: −3 + 4 = 1 ✓
4. Ex 1d: x = 4. Check: ½ × 4 + 2 = 4 ✓
5. Ex 2a: (2, 3). Check: 2 × 2 − 1 = 3 ✓ and −2 + 5 = 3 ✓
6. Ex 2b: (1, 4), so x + 3 = −2x + 6 has solution x = 1
7. Ex 2c: (2, 0)
8. Ex 2d: (−2, 0)

Easy

1. x = 2 (2x + 1 = 5)
2. x = −2 (−x + 3 = 5)
3. x = 2 (2x − 1 = 3)
4. x = −3 (x + 2 = −1)
5. x = 2 (−2x + 1 = −3)
6. x = 4 (½x + 1 = 3)
7. x = −4 (−x − 2 = 2)
8. x = −2 (3x + 1 = −5)
9. x = 2 (2x − 3 = 1)
10. x = −3 (−x + 2 = 5)
11. x = 4 (½x − 1 = 1)
12. x = −2 (3x + 2 = −4)
13. x = −2 (−2x − 1 = 3)
14. x = 4 (x − 4 = 0)

Medium

1. (1, 2)
2. m = 2, c = −1: y = 2x − 1 (11.6)
3. (2, 2)
4. (2, 0)
5. (3, 0) and (0, 3) (11.4)
6. (−2, 1)

Worded problems

1. They cross at n = 5: 20 + 25 = 45 and 10 + 35 = 45, so 5 lessons, $45
2. V = 15 when t = 5: 40 − 5 × 5 = 15 ✓, so after 5 minutes
3. 30 + 12w = 90 − 8w at w = 3: both have $66

11.8 Linear modelling (booklet p. 31)

WE DO examples (teacher)

1. Ex 1a: C: 3.50, 5.75, 8.00, 10.25
2. Ex 1b: Plot the points and rule the line from (0, 3.5)
3. Ex 1c: C = 2.25d + 3.5
4. Ex 1d: C = 2.25 × 12 + 3.5 = $30.50
5. Ex 2a: About $11.38 (C = 2.25 × 3.5 + 3.5)
6. Ex 2b: C = 2.25 × 76 + 3.5 = $174.50
7. Ex 2c: 2.25d + 3.5 = 12.05, so 2.25d = 8.55 and d = 3.8 km
8. Ex 2d: 2.25d + 3.5 = 82.25, so 2.25d = 78.75 and d = 35 km

Easy

1. $30 (C = 5n + 10)
2. 25 L (V = −5t + 40)
3. 30 km (d = 15t)
4. $16 (C = 2d + 4)
5. 10°C (T = −4t + 30)
6. $50 (C = 5n + 20)
7. $28 (P = 8h)
8. 30 L (V = −10t + 50)
9. $21 (C = 3n + 6)
10. 10 cm (H = −5t + 40)
11. 70 km (d = 20t)
12. $26 (C = 4h + 10)
13. 22.5 L (V = −15t + 60)
14. $20 (C = 2.5n + 5)

Medium

1. $20 (C = 4n + 8)
2. m = 2, c = 1: y = 2x + 1 (11.6)
3. 10 L (V = −5t + 30)
4. 30 km (d = 12t)
5. (2, 3) (11.7)
6. $20 (C = 2h + 6)

Worded problems

1. h = 30 − 2.5t; h = 30 − 12.5 = 17.5 cm; 30 − 2.5t = 0 at t = 12 hours
2. C = 20 + 0.15m; C = 20 + 18 = $38; 20 + 0.15m = 50, so m = 200 minutes
3. V = 1200 + 80t; 1200 + 80t = 5000, so 80t = 3800 and t = 47.5 minutes

11.9 Graphing parabolas (booklet p. 35)

WE DO examples (teacher)

1. Ex 1a: y: 5, 2, 1, 2, 5
2. Ex 1b: Plot (−2, 5), (−1, 2), (0, 1), (1, 2), (2, 5); join with a smooth curve
3. Ex 1c: vertex (0, 1), concave up
4. Ex 1d: $y = x^2 + 1$ ($a = 1$ is closer to 0 than $a = 3$)
5. Ex 2a: Table y: 0, 3, 4, 3, 0; vertex (0, 4), concave down
6. Ex 2b: $y = 2x^2$ ($a > 0$ and $c = 0$)
7. Ex 2c: $3^2 + 1 = 10$ ✓ Yes
8. Ex 2d: $x^2 - 9 = 0$, $x^2 = 9$, so $x = 3$ or $x = -3$

Easy

1. y: 4, 1, 0, 1, 4; vertex (0, 0), concave up
2. y: 6, 3, 2, 3, 6; vertex (0, 2), concave up
3. y: 2, −1, −2, −1, 2; vertex (0, −2), concave up
4. y: −4, −1, 0, −1, −4; vertex (0, 0), concave down
5. y: −1, 2, 3, 2, −1; vertex (0, 3), concave down
6. y: 0, −3, −4, −3, 0; vertex (0, −4), concave up
7. y: −2, 1, 2, 1, −2; vertex (0, 2), concave down
8. y: 3, 0, −1, 0, 3; vertex (0, −1), concave up
9. y: 5, −1, −3, −1, 5; vertex (0, −3), concave up
10. y: −4, 2, 4, 2, −4; vertex (0, 4), concave down
11. y: 3, 1.5, 1, 1.5, 3; vertex (0, 1), concave up
12. y: 0, 1.5, 2, 1.5, 0; vertex (0, 2), concave down
13. y: 2, −4, −6, −4, 2; vertex (0, −6), concave up
14. y: −5, −2, −1, −2, −5; vertex (0, −1), concave down

Medium

1. $y = x^2 - 3$
2. $m = 2$, $c = -1$: (0, −1) and (1, 1) (11.5)
3. $y = -x^2 + 4$
4. $y = x^2 + 1$
5. $x = 3$ (11.7)
6. $y = -x^2 - 1$

Worded problems

1. Top: $d = 0$ gives $h = 8$, so 8 m. Ground: $-0.5d^2 + 8 = 0$, $d^2 = 16$, $d = \pm 4$, so 8 m wide
2. $h = 0.25 \times 6^2 = 9$ cm
3. $y = x^2$: 4, 1, 0, 1, 4; $y = x^2 + 3$: 7, 4, 3, 4, 7. The same shape, moved up 3: vertex (0, 3)

11.10 Parabolas in real life (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: h: 20, 15, 0
2. Ex 1b: Plot (0, 20), (1, 15), (2, 0) and join with a smooth curve
3. Ex 1c: $h = -5 \times 0.5^2 + 20 = 18.75$ m
4. Ex 1d: $-5t^2 + 20 = 0$, $t^2 = 4$, $t = 2$ seconds ($t \geq 0$)
5. Ex 2a: h: 80, 75, 60, 35, 0 ($t = 0$ to 4); join with a smooth curve
6. Ex 2b: $h = -5 \times 2.5^2 + 80 = 48.75$ m
7. Ex 2c: $-5t^2 + 80 = 35$, $t^2 = 9$, $t = 3$ seconds
8. Ex 2d: At $t = 2$, $h = 60$: it has fallen $80 - 60 = 20$ m

Easy

1. $h = -5 \times 1^2 + 20 = 15$ m
2. $h = -5 \times 2^2 + 45 = 25$ m
3. $h = -5 \times 3^2 + 80 = 35$ m
4. $h = -5 \times 1^2 + 45 = 40$ m
5. $h = -5 \times 1.5^2 + 20 = 8.75$ m
6. $h = -5 \times 2^2 + 80 = 60$ m
7. $h = -5 \times 4^2 + 125 = 45$ m
8. $h = -5 \times 2.5^2 + 45 = 13.75$ m
9. $t = 2$ s ($-5 \times 2^2 + 80 = 60$)
10. $t = 1$ s ($-5 \times 1^2 + 45 = 40$)
11. $t = 3$ s ($-5 \times 3^2 + 125 = 80$)
12. $t = 1$ s ($-5 \times 1^2 + 20 = 15$)
13. $t = 2$ s ($-5 \times 2^2 + 45 = 25$)
14. $t = 4$ s ($-5 \times 4^2 + 125 = 45$)

Medium

1. $h = -5 \times 0.5^2 + 20 = 18.75$ m
2. $y = -x^2 + 2$ (11.9)
3. $h = -5 \times 1.5^2 + 45 = 33.75$ m
4. $h = -5 \times 1^2 + 80 = 75$ m
5. $m = -1$, $c = 3$: $y = -x + 3$ (11.6)
6. $h = -5 \times 3^2 + 125 = 80$ m

Worded problems

1. $-5t^2 + 20 = 0$, $t^2 = 4$, $t = 2$ s. $h = -5 \times 1.5^2 + 20 = 8.75$ m
2. Height 4 m; $-0.25d^2 + 4 = 0$, $d = \pm 4$, so 8 m wide. At $d = 1.8$: $h = 3.19$ m, which is more than 3 m, so yes
3. $s^2 = 50$, $s = \sqrt{50} = 7.1$ cm